IERG4320/IEMS5726: Data Science in Practice

Lecture 2: Basic Statistical Test

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Outline

1. Hypothesis Testing
2. Fisher's Exact Test
3. Chi-squared Test (χ^2 Test)
4. Wilcoxon Rank-Sum Test (Mann–Whitney U test)
5. Wilcoxon Signed-Rank Test

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Why statistics is needed in data science?

Data scientists use statistics to **gather**, **review**, **analyze**, and **draw conclusions** from data, as well as apply quantified mathematical models to appropriate variables

- **Population**: the source of data to be collected.
- **Sample**: a portion of the population.
- **Variable**: any data item that can be measured or counted.
- **Quantitative analysis (statistical)**: collecting and interpreting data with patterns and data visualization.
- **Qualitative analysis (non-statistical)**: producing generic information from other non-data forms of media.

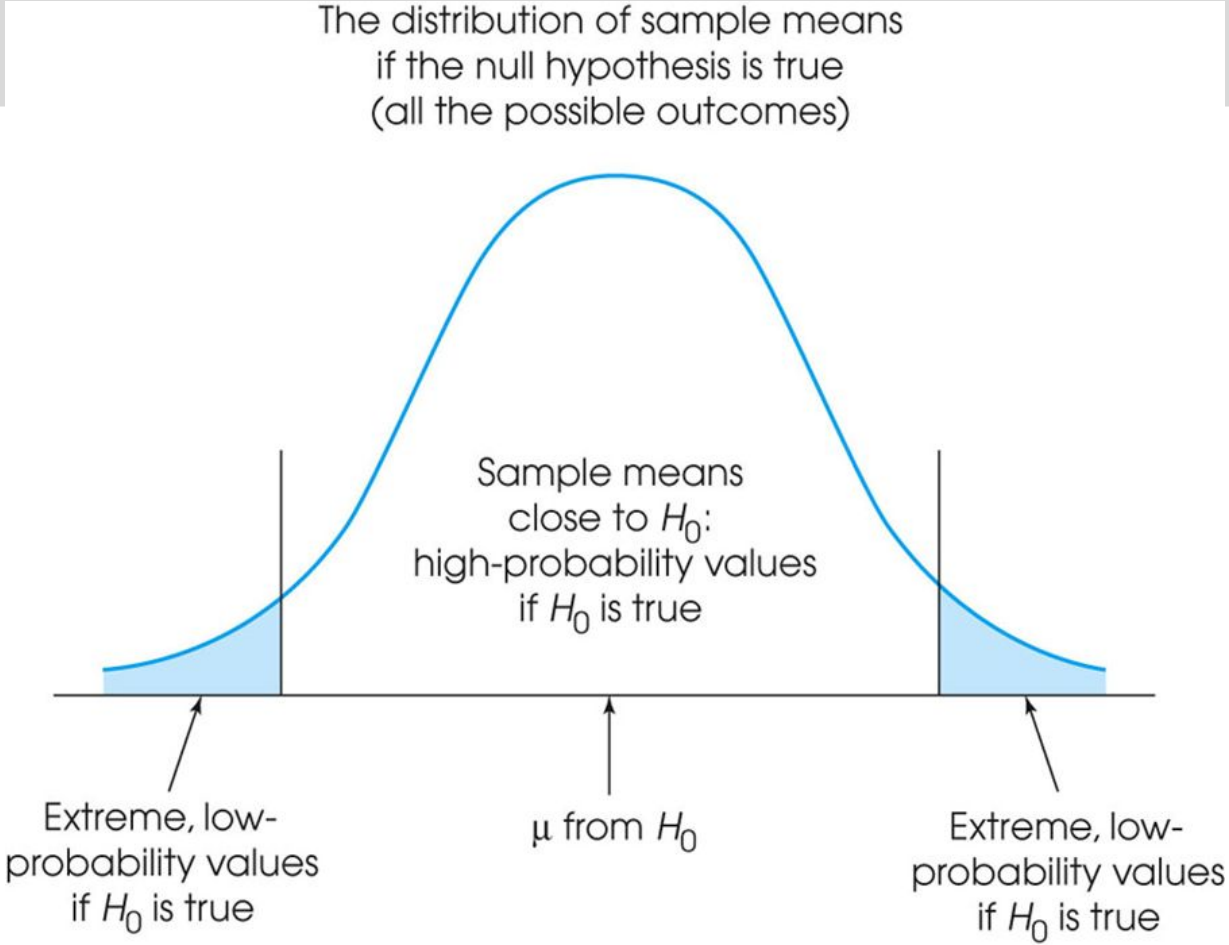
Why statistics is needed in data science?

- **Descriptive statistics**: characteristics of a population.
- **Inferential statistics**: predictions for a population.
- **Central tendency (measures of the center)**: mean (average of all values), median (central value of a data set), and mode (the most recurrent value in a data set).
- **Measures of the spread**:
  - **Range**: the distance between each value in a data set.
  - **Variance**: the distance between a variable and its expected value.
  - **Standard deviation**: the dispersion of a data set from the mean.

Reference: https://datascience.virginia.edu/news/how-much-do-data-scientists-need-know-about-statistics

Hypothesis Testing

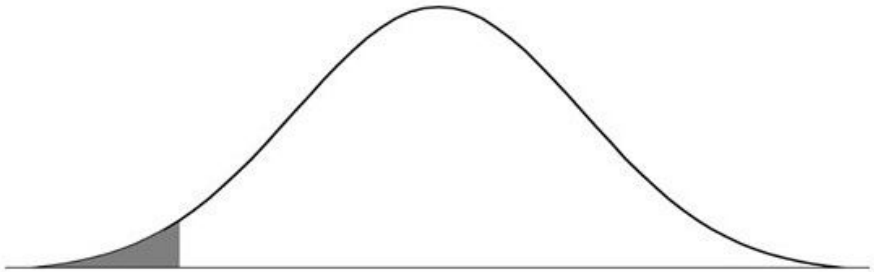
1. The general goal of a hypothesis test is to **rule out chance (sampling error)** as a plausible explanation for the results from a research study
2. Hypothesis testing is a technique to help determine whether a specific treatment has an effect on the individuals in a population
3. If the individuals in the sample are noticeably different from the individuals in the original population, we have evidence that the treatment has an effect
4. However, it is also possible that the difference between the sample and the population is simply sampling error


The distribution of sample means
if the null hypothesis is true
(all the possible outcomes)
Sample means
close to H_0:
high-probability values
if H_0 is true
Extreme, low-
probability values
if H_0 is true
μ from H_0
Extreme, low-
probability values
if H_0 is true

Hypothesis Testing (Steps)

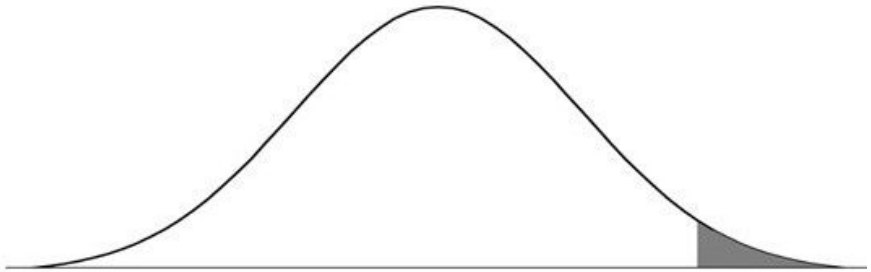
1. **State the Hypothesis**
   - **Null hypothesis (H_0)**: phenomenon being studies produces no effect or make no difference
   - **Alternative hypothesis (H_a)**: phenomenon to be tested or make differences
2. **Choose the significant level (α)**: Common choices, 0.05, 0.01 or 0.1
3. **Select the appropriate test**: e.g. Z-test, t-test, Chi-square test, … etc
4. **Calculate the test statistic**: The most common one, z-score, a ratio comparing the obtained difference between the sample mean and the hypothesized population mean versus the standard error
5. **Determine critical region**: defined by sample means that are almost impossible to obtain if the phenomenon has no effect
6. **Make a decision**: Reject H_0 or fail to reject H_0
   - A large value for the test statistic shows that the obtained mean difference is more than would be expected if there is no treatment effect

One-tailed Critical Regions



$$H_0 : p \geq p_0 \quad \text{and} \quad H_A : p < p_0$$

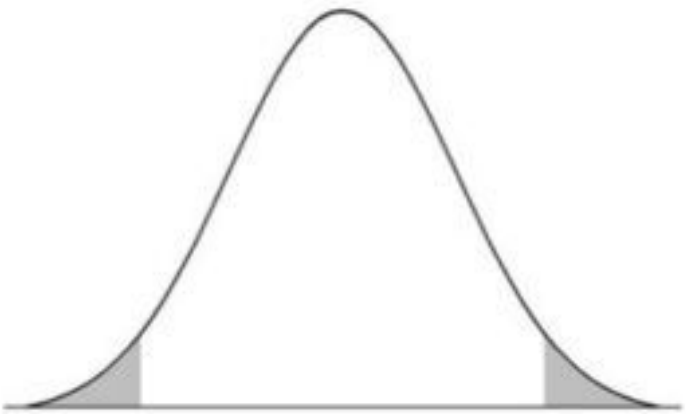
Reject H_0 if $z < -z_\alpha$



$$H_0 : p \leq p_0 \quad \text{and} \quad H_A : p > p_0$$

Reject H_0 if $z > -z_\alpha$

Two-tailed Critical Regions



$$H_0 : |p| \leq p_0 \text{ and } H_A : |p| > p_0$$

Reject H_0 if $|z| > z_\alpha$

p-value

- The probability that a particular statistical measure, such as the mean or standard deviation, of an assumed probability distribution will be greater than or equal to (or less than or equal to in some instances) observed results.
- i.e. p-value = **the area of the critical region(s)**
- E.g. Suppose we test whether a coin is fair (H_0: p=0.5) vs. biased (H_a: p≠0.5)
  - If we flip 100 times and get 60 heads, the p-value is the probability of getting ≥ 60 or ≤ 40 heads if H_0 is true.
  - p-value = 2 P(X ≥ 60) = 2(0.0287) = 0.0574 > 0.05

$$P(X \geq 60) = \sum_{k=60}^{100} \binom{100}{k} (0.5)^k (0.5)^{100-k} = \sum_{k=60}^{100} \binom{100}{k} (0.5)^{100}$$

  - We cannot reject H_0 under 95% significance level

Fisher’s Exact Test

Fisher's Exact Test

- A statistical test used in the analysis of 2×2 contingency tables
- A contingency tables displays the frequency distribution of the variables
- Example:

Rolling 2 dice	Experiment	Theory
Sum ≥ 10	15	6
Sum < 10	57	30

- In general:

	Group I	Group II
Scene 1	*a*	*b*
Not scene 1	*c*	*d*

$$p\text{-value} = \sum_{i=0}^{\min(b,c)} \frac{\binom{a+b}{a+i}\binom{c+d}{c-i}}{\binom{a+b+c+d}{a+c}}$$

Example: Fisher's Exact Test

Question: Based on the observation, are the dice fair? H_0: The dice is not fair

Rolling 2 dice	Experiment	Theory
Sum ≥ 10	15	6
Sum < 10	57	30

$$p\text{-value} = \sum_{i=0}^{6} \frac{\binom{15+57}{15+i}\binom{6+30}{6-i}}{\binom{15+57+6+30}{15+6}} = 0.4051$$

Conclusion: Failed to reject H_0

```
import scipy.stats as stats
oddsratio, pvalue = stats.fisher_exact([[15,57],[6,30]], "greater")
print(oddsratio, " ", pvalue)
```

```
1.3157894736842106   0.4051066656100187
```

More about Fisher’s Exact Test

- Advantages of Fisher’s Exact Test

1. Ideal for small sample sizes
2. Analyzes two binary variables
3. Tests the null hypothesis that the two variables are independent (no association)
4. Evaluate the exact probability

- Disadvantages

1. Not all the variables are binary
2. Evaluation of the exact probability takes time

Chi-squared Test (χ^2 Test)

Chi-squared Test (χ^2 Test)

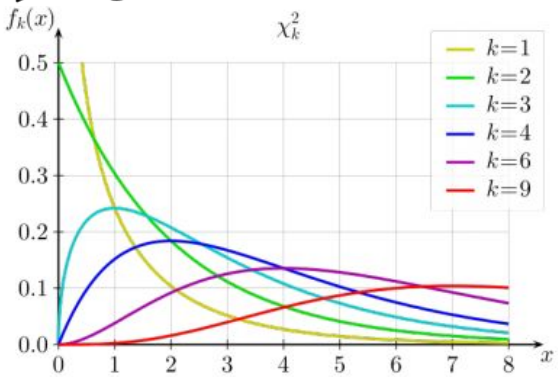
- A statistical test to determine where there is a statistically significant difference between the expected frequencies and the observed frequencies in one or more categories of a contingency table

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = N \sum_{i=1}^{n} \frac{(O_i/N - p_i)^2}{p_i}$$

- χ^2: Pearson's cumulative test statistic, chi-square distribution
- O_i: number of observation of type i
- $E_i = Np_i$: expected count of type i under null hypothesis
- N: total number of observations

Chi-square Distribution

- Define $X_i \sim N(0, 1)$
- Define $Y = X_1^2 + X_2^2 + \ldots + X_k^2$
- Y is chi-square distribution with (parameter) k degrees of freedom
- pdf: $$f_X(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, x > 0$$
- CDF: $$F(x) = \frac{1}{\Gamma(k/2)} \gamma(\frac{k}{2}, \frac{x}{2}), x > 0$$
- Properties: $$E[X] = k$$ $$Var[X] = 2k$$



Example: Chi-squared Test

A 6-sided dice is thrown 60 times. The number of times it lands with 1, 2, 3, 4, 5 and 6 face up is 5, 8, 9, 8, 10 and 20, respectively. Is the dice biased, according to the Pearson's chi-squared test at a significance level of 95% and/or 99%?

i	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2/E_i$
1	5	10	−5	25	2.5
2	8	10	−2	4	0.4
3	9	10	−1	1	0.1
4	8	10	−2	4	0.4
5	10	10	0	0	0
6	20	10	10	100	10
				Sum	13.4

Reference: Wikipedia

Example: Chi-squared Test (Continue)

```
from scipy.stats import chisquare
chi2, pvalue = chisquare([5,8,9,8,10,20])
print(chi2, " ", pvalue)
```

```
13.4   0.01990522033477438
```

- H_0: $\chi^2 = 0$, the die is fair
- Degree of freedom = 5
- $\chi^2 = 13.4$
- By table lookup,
  - $P(\chi^2 > 11.070, df = 5) = 0.05$
    → If we decide to use 95% significance level, reject H_0
    → The die is not fair
  - $P(\chi^2 > 15.086, df = 5) = 0.01$
    → If we decide to use 99% significance level, cannot reject H_0
    → No conclusions can be made here
- If you look for the exact value of $P(\chi^2 > 13.4, df = 5)$, programmatically, it is 0.0199

Wilcoxon Tests

Wilcoxon Tests

- Wilcoxon tests are non-parametric alternatives to t-tests when data does not meet normality assumptions.

1. Wilcoxon Rank-Sum Test (Mann-Whitney U Test) (for independent samples)
2. Wilcoxon Signed-Rank Test (for paired/dependent samples)

Wilcoxon Rank-Sum Test (Mann–Whitney U test)

- X and Y are random variables
- H_0: Randomly selected values X and Y from two populations, the probability of X being greater than Y is equal to the probability of Y being greater than X

- Assumptions:
  - All the observations from both groups are **independent** of each other
  - Under null hypothesis H_0, the distributions of both populations are equal
  - The alternative hypothesis H_1, the distributions are not equal.

Wilcoxon Rank-Sum Test Procedures

1. Rank the data of both the groups in ascending order
   - If any values are equal average their ranks
2. Add up the ranks in either group
   - T = sum of ranks of a group
3. Calculate U statistics as T - n(n+1)/2, where n is the same of the corresponding group
4. Use normal approximation to evaluate the p-value

$$z = \frac{U - m_U}{\sigma_U}, \quad m_U = \frac{n_1 n_2}{2}, \quad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}.$$

Example: Wilcoxon Rank-Sum Test

X	r_X	Y	r_Y
3.99	27	3.18	7
3.79	24	2.84	5
3.60	18	2.90	6
3.73	22	3.27	11
3.21	8	3.85	26
3.60	18	3.52	14
4.08	28	3.23	9
3.61	20	2.76	4
3.83	25	3.60	18
3.31	12	3.75	23
4.13	29	3.59	16
3.26	10	3.63	21
3.54	15	2.38	2
3.51	13	2.34	1
2.71	3		
	Sum=272		**Sum=163**

$U_X = 272 - 15(15+1)/2 = 152$

$U_Y = 163 - 14(14+1)/2 = 58$

$m_U = 105$

$\sigma_U = 56.1249$

$z = (152 - 105) / 56.1249$

$= 0.8374$

p-value = 0.04024

We cannot reject H_0 at 99% significant level

Example: Wilcoxon Rank-Sum Test (Python)

```python
from scipy.stats import ranksums, mannwhitneyu
import numpy as np
X = np.array([3.99, 3.79, 3.60, 3.73, 3.21, 3.60, 4.08, 3.61, 3.83, 3.31, 4.13, 3.26, 3.54, 3.51, 2.71])
Y = np.array([3.18, 2.84, 2.90, 3.27, 3.85, 3.52, 3.23, 2.76, 3.60, 3.75, 3.59, 3.63, 2.38, 2.34])
U, pvalue = ranksums(X,Y)
print(U, pvalue)

statistic, pvalue = mannwhitneyu(X,Y)
print(statistic, pvalue)
```

```
2.0512481682183283 0.04024278524556949
152.0 0.042313171142852365
```

Wilcoxon Signed-Rank Test

- It is an alternative to the paired Student's t-test
- H_0: Two dependent samples were selected from populations having the same distribution

- Assumptions:
  - Data are paired and come from the same population
  - Each pair is chosen randomly and independently

Wilcoxon Signed-Rank Test Procedures

1. Calculate the differences
2. Exclude any differences which are zero
3. Count the ranks based on the magnitude of the difference
4. Sum the signed-rank statistics as $W = \sum_{i=1}^{N_r} [\mathrm{sgn}(x_{2,i} - x_{1,i}) \cdot R_i]$
5. Test statistics is $z = \frac{W}{\sigma_W}$, where $\sigma_W = \sqrt{\frac{N_r(N_r+1)(2N_r+1)}{6}}$
6. Evaluate the p-value based on *t*-distribution

Example: Wilcoxon Signed-Rank Test (W-statistic)

X	Y	X - Y	Rank
6.1	5.2	0.9	3.5
7.0	7.9	-0.9	3.5
8.2	3.9	4.3	10
7.6	4.7	2.9	7
6.5	5.3	1.2	5
8.4	5.4	3.0	8
6.9	4.2	2.7	6
6.7	6.1	0.6	2
7.4	3.8	3.6	9
5.8	6.3	-0.5	1

W = 3.5+(-3.5)+10+7+5+8+6
+2+9+(-1)
= 46

σ_W = sqrt(10*11*21/6)
= 19.6214

df = 10 - 1 = 9

P(| t_9 | > 46/19.6214, df = 9)
= (1-0.9781)*2 = 0.0438

We cannot reject H_0 at 99% significant level

Example: Wilcoxon Signed-Rank Test (Python)

- However, `wilcoxon()` in python uses **T-statistic** to calculate the p-value, which is different from the previous slide
- T-statistic is based on t-distribution, while W-statistics is based on Wilcoxon table

```python
from scipy.stats import wilcoxon
import numpy as np
X = np.array([6.1, 7.0, 8.2, 7.6, 6.5, 8.4, 6.9, 6.7, 7.4, 5.8])
Y = np.array([5.2, 7.9, 3.9, 4.7, 5.3, 5.4, 4.2, 6.1, 3.8, 6.3])
T, pvalue = wilcoxon(X, Y)
print(T, pvalue)

d = np.array([9, -9, 43, 29, 12, 30, 27, 6, 36, -5])
T, pvalue = wilcoxon(X, Y)
print(T, pvalue)
```

```
5.0 0.01953125
5.0 0.01953125
```

Some Other Statistical Tests

1. **T-tests**, **Paired t-tests**: To compare two variables from the same population.
2. **Analysis of Variance (ANOVA)**: The t-tests described above have several limitations.
3. **Correlation**: To describe the degree to which one variable is linearly related to another.
4. **Regression**: To develop an estimating equation that relates the known variables to the unknown variable .
5. **Non-parametric Tests**: Experiments that do not require the underlying population for assumptions.

- We may cover some of them later in the course, no worries.